

This exam paper is part of a set used to assess student abilities as follows:

		Question
CLO 1	Explain and use the theory of sets, real number system and relations.	Q1-Q9
CLO 2	Understand and analyze mathematical contents related to sets, real number system and relations.	Q1-Q9
CLO 3	Enable to write the steps to solve mathematical problems.	Q1-Q9
CLO 4	Enable to use mathematical techniques to solve mathematical problems.	Q1-Q9
CLO 5	Have moral and ethical values, respect rights, listen to others' opinions, and able to work collaboratively with others.	-

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Question 1.

1.1 Given $A = \{4, 6, 8, 9\}$, $A \cup B = \{4, 6, 7, 8, 9, 11\}$, and $A \cap B = \{4, 8\}$.

Find B .

(2 points)

Given a universal set U , set A and set B as follows:

$$U = \{x \in I^+ \mid x < 10\},$$

$$A = \{x \in U \mid x \text{ is even}\}, \text{ and}$$

$$B = \{x \in U \mid x^2 < 36\}.$$

With these sets, find the solutions for 1.2 - 1.5.

(2 points each)

1.2 $A \cup B$.

1.3 $B - A$.

1.4 $(A - B)'$.

1.5 $P(A \cap B)$.

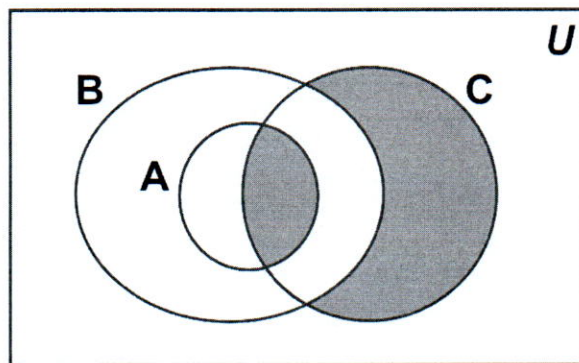
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Question 2. Given non-empty sets A , B , and C in a universal set U .

2.1 If $B \subset C$ and $A \cap C = \emptyset$, find $(A \cup B) \cap C$. (2 points)

2.2 Show that $A \cap (B - C)' = (A \cap B') \cup (A \cap C)$. (2 points)

2.3 Given the following Venn-Euler diagram depicting sets A , B , and C in a universal set U :



Write a set notation that represents the shaded region using set operations and the given set labels. (3 points)

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Question 3. In a school with 60 students, there are three clubs:

A: Science Club, B: Art Club, and C: Sports Club.

Every student joins at least one club.

A survey found that:

34 students are members of the Science Club,
30 students are members of the Art Club,
8 students are in both Science and Art Clubs,
23 students are in both Science and Sports Clubs,
20 students are in both Art and Sports Clubs, and
6 students are only in the Science Club.

3.1 Find the number of students who are members of all 3 clubs. (5 points)

3.2 Find the number of students who are members of only one club. (5 points)

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Question 4.

4.1 Convert the repeated decimal $0.4\overline{9763}$ to a fraction.

(2 points)

4.2 Determine the interval of $x^2 - y$ when $-6 < x < 5$ and $-2 < y < 6$. (3 points)

4.3 Find the remainder of the expression $\frac{3x^3 - 15x^2 - 9x + 1}{3x + 1}$. (3 points)

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Question 5.

Define the binary operator $\otimes$ by $x \otimes y = x + y + 2$ where x and y are integers.

5.1 Evaluate $(1 \otimes 3) \otimes 4$.

(3 points)

5.2 Find the identity of the operator $\otimes$.

(3 points)

5.3 Find the inverse of 4 for the operator $\otimes$.

(3 points)

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Question 6.

6.1 Solve the following inequality:

$$x(x+2)(1-x)(x-3) > 0.$$

(4 points)

6.2 Find the solution of the following inequality:

$$\frac{x^3 + 4x^2 - 3x - 18}{x^2 - 3x + 2} \geq 0.$$

(7 points)

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Question 7. Given $A = \{1, 2, 3\}$ and $B = \{b, \phi\}$.

7.1 Find $A \times B$.

(3 points)

7.2 Find the total number of relations from A to B .

(2 points)

7.3 Let r be a relation from A to B such that $n(r) = 3$ and $2 \notin D_r$.

Give one example of the relation r .

(2 points)

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Question 8. Given the following relations:

$$r_1 = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y \leq x + 1\} \text{ and } r_2 = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y \geq -x\}.$$

8.1 Sketch the graph of r_1 .

(3 points)

8.2 Let $r = r_1 \cap r_2$. Find D_r and R_r .

(5 points)

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Question 9.

9.1 Let $r^{-1} = \{(1, \phi), (2, \phi), (1, a), (1, b)\}$ be the inverse of r .

Find the relation r .

(2 points)

9.2 Given the relation $r = \left\{ (x, y) \in \mathbb{R} \times \mathbb{R} \mid y = \frac{3x+2}{x-5} \right\}$.

Find r^{-1} and $D_{r^{-1}} \cap R_{r^{-1}}$.

(6 points)

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